YiRage (YPK) — 系统架构 · RuntimeFusion 数据流

Inference engine owns scheduling/KV; RF owns fusion identity; search and execute use the same backend

主数据流　　虚线框 = 架构分区

回退 / 反馈　　盒内 = 模块 + 职责

项目介绍

YiRage 是面向大模型推理的多后端算子编译与运行时系统，核心能力是计算图超优化和 RuntimeFusion。

系统离线搜索并验证后端特定融合计划，再将 FusionCapsule 注入 vLLM 或 SGLang 的逐步解码流程。

INFERENCE ENGINE · 调度与 KV 所有者

vLLM
continuous batch · PagedAttention

SGLang
RadixAttention · extend meta

Decode Metadata
block table · seq lens · SM budget

RUNTIMEFUSION · 每个 decode step

RF.step(inputs, meta)
select 0..N capsules

FusionCapsule
MLP gate_up + down
segment pipeline

Native Fallback
skip → engine native ops

SEARCH & PLAN · offline / warm-up

superoptimize(backend)
KN / TB graph search

FusionPlan / μGraph
profile · fingerprint verify

Persistent Cache
shape bucket restore

Ray Coordinator
optional distributed search

fallback / re-search

BACKEND MANAGER · 编译与执行边界

BackendRegistry
search + execute same backend

KNGraph / TBGraph
transpiler → kernels

yirage.core
Cython runtime

BACKENDS → HARDWARE

CUDA / MACA
NVIDIA GPU · MetaX C500/C550

ROCm / MPS
AMD GPU · Apple Silicon

Ascend / NKI
Huawei 910 · AWS Neuron

CPU / MLIR / Triton
x86 · ARM · extra targets

主链：Engine metadata → RF.step → FusionCapsule → BackendRegistry → target hardware。